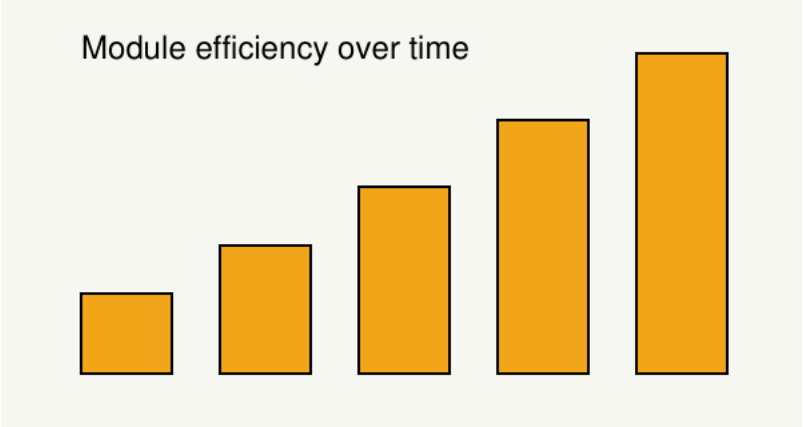
Solar Energy: A Practical Overview

This sample document exists to exercise the **.mvpdf** converter. It mixes headings, body text, an image and both external and internal links. Read more at the U.S. Department of Energy or jump to the section on storage.

How Photovoltaic Cells Work

Solar panels convert sunlight directly into electricity using photovoltaic cells. Each cell is a sandwich of doped silicon layers. When photons strike the cell they knock electrons loose, and the electric field at the junction pushes those electrons into a current that flows through the attached circuit.



Panel efficiency has climbed steadily. Commercial modules now convert a little over a fifth of incoming sunlight into electricity, while laboratory cells built from stacked materials have passed forty percent. See the NREL efficiency chart for details.

Costs and Manufacturing

The price of solar modules has fallen by roughly ninety percent over the last decade. Cheaper manufacturing, larger factories and better wafer cutting all contributed. In many regions new solar farms are now the least expensive source of new electricity generation.

Installation labour, permitting and grid connection now make up a growing share of the total cost of a rooftop system, which is why so-called soft costs are a focus for policy.

Storage and the Grid

Grid operators must balance supply and demand every second. Because solar output peaks at midday and drops to zero at night, batteries are increasingly paired with solar farms to shift energy into the evening peak. Lithium-ion packs dominate today, while flow batteries and thermal storage compete for longer durations.

During a grid outage, a home battery with an islanding inverter can keep essential circuits running. Without that equipment, rooftop panels shut down for the safety of line workers.

End of Life

Recycling programs are growing so that glass, metal, and silicon can be recovered when old panels are retired. Most panels carry twenty-five year warranties, so the first large wave of retirements is only now arriving.

Back to the top